



COURSE OUTLINE

Course Title: Data Structures and Algorithm Analysis

Course Code: Cosc3111

Credit Hours: 3

ECTS: 5

Prerequisite(s): CoSc2111

Course objective:

Upon completion of the course students should be able to:

On completion of this course students should be able to:

- To introduce the most common data structures like stack, queue, linked list
- To give alternate methods of data organization and representation
- To enable students use the concepts related to Data Structures and Algorithms to solve real world problems
- To practice Recursion, Sorting, and searching on the different data structures
- To implement the data structures with a chosen programming language

Course Description:

This course focuses on the study of data structures, algorithms and program efficiency. Topics include: analysis of time and space requirements of algorithms; program efficiency improving techniques, abstract data types such as linked lists, stacks, queues, trees (traversal, implementations); simple searching algorithms (linear search, binary search, ...), simple sorting algorithms (bubble sort, insertion sort, selection sort, ...), advanced sorting algorithms (merge sort, quick sort, heap sort ...)

Course content:

1. Introduction to Data Structures
 - 1.1. Introduction
 - 1.2. Abstract Data Type and Abstraction
2. Algorithm and Algorithm analysis
 - 2.1. Analysis
 - 2.2. Properties of Algorithm
 - 2.3. Analysis of Algorithm
3. Simple Sorting and Searching Algorithms
 - 3.1. Sorting
 - 3.1.1. Selection Sort
 - 3.1.2. Bubble Sort
 - 3.1.3. Insertion Sort
 - 3.1.4. Pointer Sort
 - 3.2. Searching
 - 3.2.1. Linear/Sequential Searching



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3.3.3.2.2.Binary Searching

4. Linked Lists
 - 4.1. Review on Pointer, Dynamic Memory allocation and De-allocation
 - 4.2. Singly Linked Lists
 - 4.3. Doubly Linked Lists
5. Stacks and Queues
 - 5.1. Basic Stack Operations
 - 5.2. Basic Queue Operations
 - 5.3. Implementation of Stacks and queues
6. Tree Structures
 - 6.1. Binary Trees and Binary Search Trees
 - 6.2. Basic Tree Operations
 - 6.3. Traversing in a Binary Tree
 - 6.4. General Trees and Their Implementations
7. Graphs
 - 7.1. Introduction
 - 7.2. Describing graphs
 - 7.3. Directed Graphs
 - 7.4. Traversing a graph
8. Advanced Sorting and Searching
 - 8.1. Sorting
 - 8.1.1 Heap Sort
 - 8.1.2.Quick Sort
 - 8.1.3.Merge Sort
 - 8.1.4.Shell Sort
 - 8.2. Advanced Searching
 - 8.2.1.Hashing

Teaching & Learning Methods: Lectures, Individual and group Assignments

Assessment/Evaluation & Grading System:

- Continuous Assessment -----60%
- Final Exam -----40%

Text books & References:

- Robert Lafore, “Data Structures and Algorithms in JAVA, 2nd Ed.”, Sams Publishing
- Jean Paul Tremblay, Paul G. Soreson, “An Introduction to Data Structures with Applications”, Mc. Graw Hill Computer Science Series
- E. Horowitz, S.Sahni and Dinesh Mehta. Fundamentals of data structures in C++, W.H Freeman and Company (1995)
- Sanjay Pahuja, A practical approach to data structures and algorithms, New age International publishers, 2008